7

Entropy

Learning Objectives

After reading this chapter you should be able to

LO 7.1	Describe three E's of thermodynamics
LO 7.2	Formulate equation of inequality of Clausius
LO 7.3	Summarize entropy concept by utilizing Clausius' theorem principles
LO 7.4	Represent reversible processes and cycle on temperature entropy plot
LO 7.5	Discuss the entropy change in any process
LO 7.6	Explain the principle of increase of entropy
LO 7.7	Describe the various applications of entropy principle in quantifying irreversibility of the process
LO 7.8	Analyzing entropy transfer with flow of heat and not with work
LO 7.9	Examine possible ways of entropy increase in closed and open system
LO 7.10	Express combined consequences of First and Second laws
LO 7.11	Obtain reversible work done in a steady flow system
LO 7.12	Express entropy and its dependence on direction
LO 7.13	Correlate entropy with disorderness
LO 7.14	Discuss the concepts of absolute entropy and postulatory thermodynamics

7.1 INTRODUCTION

LO 7.1
Describe three E's of thermodynamics

The first law of thermodynamics was stated in terms of cycles first and it was shown that the cyclic integral of heat is equal to the cyclic integral of work. When the first law was applied for thermodynamic processes, the existence of a property, the internal energy, was found. Similarly, the second law was also first stated in terms of cycles executed by systems. When applied to processes, the second law also leads to the definition of a new property, known as entropy. If the first law is said to be the law of internal energy, then the second law may be stated to be the law of entropy. In fact, *thermodynamics is the study of three E's, namely, energy, equilibrium and entropy.*

7.2 THE INEQUALITY OF CLAUSIUS

LO 7.2 Formulate equation of inequality of Clausius

Let us consider two heat engine cycles, one reversible (E_R) and the other irreversible (E_I), operating between the same two constant temperature reservoirs at temperatures T_1 and T_2 as shown in Fig. 7.1.

Our objective is to find the expression of cyclic integral of $đQ$ and $đQ/T$ both for reversible and irreversible heat engines, where T is the temperature of system boundary over which heat transfer takes place.

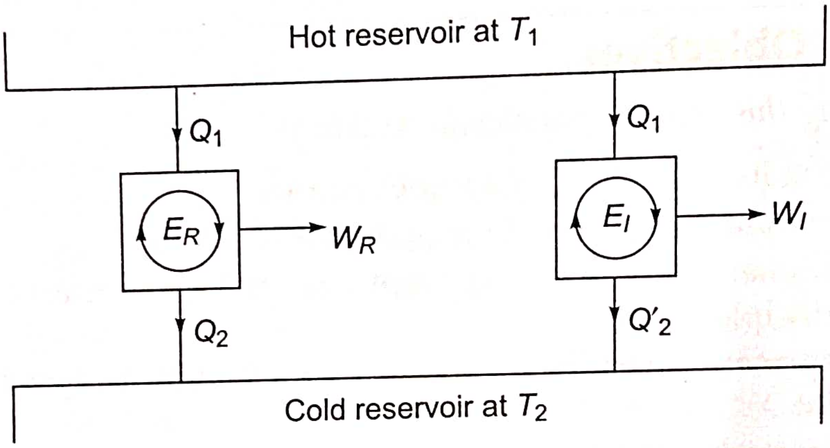

Fig. 7.1 *Two heat engines, one reversible and other irreversible operating between same two reservoirs*

From Kelvin-Planck statement of second law of thermodynamics, both for reversible and irreversible heat engines, we have

$$\oint đQ|_{HE} = Q_1 - Q_2 \ (\text{or} = Q_1 - Q_2') > 0 \tag{7.1}$$

For the reversible heat engine,

$$\oint \frac{đQ}{T}\bigg|_{\text{Rev, HE}} = \frac{Q_1}{T_1} - \frac{Q_2}{T_2} = 0 \tag{7.2}$$

Since the efficiency of the irreversible engine η_I is less than that of the reversible engine η_R working between the same two reservoirs, we have

$$\eta_I < \eta_R$$

or

$$\frac{W_I}{Q_1} < \frac{W_R}{Q_1}$$

For the supply of the same quantity of heat Q_1 to both the engines, we get

$$W_I < W_R$$

or

$$Q_1 - Q'_2 < Q_1 - Q_2$$

$\therefore$

$$Q'_2 > Q_2$$

For the irreversible heat engine

$$\oint \frac{đQ}{T}\bigg|_{\text{Irrev,HE}} = \frac{Q_1}{T_1} - \frac{Q_2'}{T_2} < 0 \tag{7.3}$$

So, in general, we have

$$\oint \frac{đQ}{T}\bigg|_{\text{HE}} \leq 0 \tag{7.4}$$

Next, we consider two refrigerator/heat pump cycles, one reversible (Ref/HP$_R$) and the other irreversible (Ref/HP$_I$), operating between the same two constant temperature reservoirs at temperatures T_1 and T_2 as shown in Fig. 7.2.

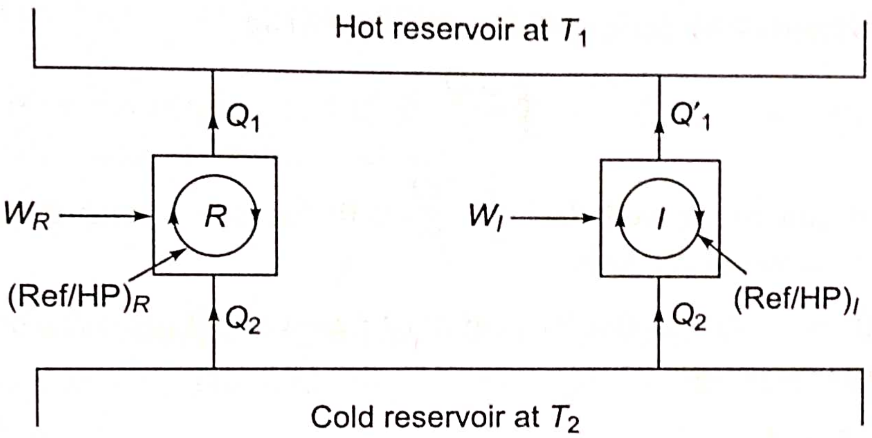

Fig. 7.2 *Two refrigerators/heat pumps, one reversible and other irreversible operating between same two reservoirs*

From Clausius statement of second law of thermodynamics, both for reversible and irreversible heat pumps/refrigerators, we have

$$\oint đQ\big|_{\text{Ref/HP}} = Q_2 - Q_1 \;(\text{or} = Q_2 - Q_1') < 0 \tag{7.5}$$

For the reversible heat pump/refrigerator,

$$\oint \frac{đQ}{T}\bigg|_{\text{Rev,Ref/HP}} = \frac{Q_2}{T_2} - \frac{Q_1}{T_1} = 0 \tag{7.6}$$

Since for the same Q_2 and working between same two reservoirs, the irreversible refrigerator/ heat pump requires more work input than that of reversible one, we have

$$W_I > W_R$$

or $$W_I + Q_2 > W_R + Q_2$$

or $$Q'_1 > Q_1$$

For the irreversible heat pump/refrigerator,

$$\oint \frac{đQ}{T}\bigg|_{\text{Irrev,Ref/HP}} = \frac{Q_2}{T_2} - \frac{Q_1'}{T_1} < 0 \tag{7.7}$$

So, in general, we have

$$\oint \frac{đQ}{T}\bigg|_{\text{Ref/HP}} \leq 0 \tag{7.8}$$

From the above discussion, important conclusions that can be drawn that although the sign of $\oint đQ$ differs, depending on whether it is heat engine or refrigerator/heat pump, $\oint \frac{đQ}{T} \leq 0$ for all cyclic devices.

Using Eqs. (7.2) and (7.6), for any reversible cycle, one can write

$$\oint_R \frac{đQ}{T} = 0 \tag{7.9}$$

It implies that the cyclic integral of $đQ/T$ for a reversible cycle is equal to zero. This is known as *Clausius' theorem*. The letter R emphasizes the fact that the equation is valid only for a reversible cycle.

Using Eqs. (7.4) and (7.8), for any cycle, one can write

$$\oint \frac{đQ}{T} \leq 0 \tag{7.10}$$

Similarly, it can be proved that $\oint \frac{đQ}{T} \leq 0$ for any cyclic device operating between more than two reservoirs.

Equation (7.10) is known as the *inequality of Clausius. It provides the criterion of the reversibility of a cycle,*

If $\oint \frac{đQ}{T} = 0$, the cycle is reversible.

$\oint \frac{đQ}{T} < 0$, the cycle is irreversible and possible.

$\oint \frac{đQ}{T} > 0$, the cycle is impossible, since it violates the second law.

7.3 THE PROPERTY OF ENTROPY

LO 7.3 Summarize entropy concept by utilizing Clausius' theorem principles

Let a system be taken from an initial equilibrium state i to a final equilibrium state f by following the reversible path R_1 (Fig. 7.3). The system is brought back from f to i by following another reversible path R_2. Then the two paths R_1 and R_2 together constitute a reversible cycle. From Clausius' theorem

$$\oint_{R_1 R_2} \frac{đQ}{T} = 0$$

The above integral may be replaced as the sum of two integrals, one for path R_1 and the other for path R_2

$$\int_{\substack{i \\ R_1}}^{f} \frac{đQ}{T} + \int_{\substack{f \\ R_2}}^{i} \frac{đQ}{T} = 0$$

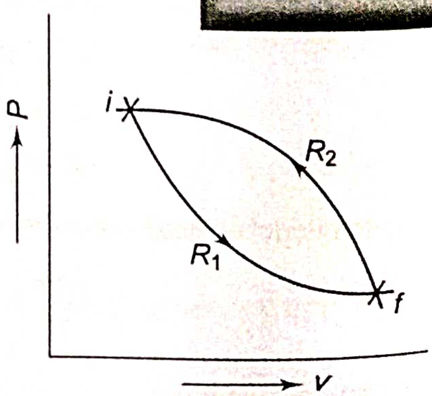

Fig. 7.3 *Two reversible paths R_1 and R_2 between two equilibrium states i and f*

or

$$\int_{i\,R_1}^{f} \frac{đQ}{T} = -\int_{f\,R_2}^{i} \frac{đQ}{T}$$

Since R_2 is a reversible path

$$\int_{i\,R_1}^{f} \frac{đQ}{T} = \int_{i\,R_2}^{f} \frac{đQ}{T}$$

Since R_1 and R_2 represent any two reversible paths, $\int_{i\,R}^{f} \frac{đQ}{T}$ is independent of the reversible path connecting i and f. Therefore, there exists a property of a system whose value at the final state f minus its value at the initial state i is equal to $\int_{i\,R}^{f} \frac{đQ}{T}$. This property is called entropy, and is denoted by S. If S_i is the entropy at the initial state i, and S_f is the entropy at the final state f, then

$$\int_{i\,R}^{f} \frac{đQ}{T} = S_f - S_i \tag{7.11}$$

Note that we are concerned with the changes in entropy during a process undergone by a system instead of its absolute value. Accordingly we have defined the change in entropy by Eq. (7.11)

When the two equilibrium states are infinitesimally near

$$\frac{đQ_R}{T} = dS \tag{7.12}$$

where dS is an exact *differential* because S is a point function and a property. The subscript R in $đQ$ indicates that heat $đQ$ is transferred *reversibly*.

The word 'entropy' was first used by Clausius, taken from the Greek word '*tropee*' meaning 'transformation'. It is an extensive property, and has the unit J/K. The specific entropy

$$s = \frac{S}{m} \text{ J/kg K}$$

If the system is taken from an initial equilibrium state i to a final equilibrium state f by an *irreversible path*, since entropy is a point or state function, and the entropy change is independent of the path followed, the non-reversible path is to be replaced by a reversible path to integrate for the evaluation of entropy change in the irreversible process (Fig. 7.4).

$$S_f - S_i = \int_i^f \frac{đQ_{\text{rev}}}{T} = (\Delta S)_{\text{irrev path}} \tag{7.13}$$

Integration can be performed only on a reversible path.

Thus the entropy change in any irreversible or real process connecting two equilibrium points, the initial and the final, can be estimated by replacing a reversible path between the same points.

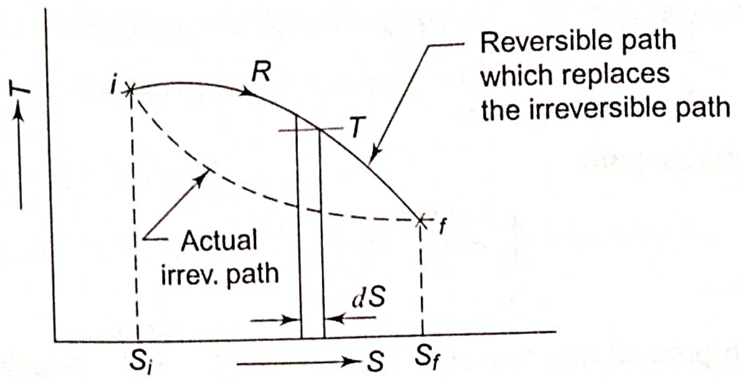

Fig. 7.4 *Integration can be done only on a reversible path*

7.4 TEMPERATURE-ENTROPY PLOT

LO 7.4 Represent reversible processes and cycle on temperature entropy plot

The infinitesimal change in entropy dS due to reversible heat transfer $đQ$ at temperature T is

$$dS = \frac{đQ_{\text{rev}}}{T}$$

If $đQ_{\text{rev}} = 0$, i.e. the process is reversible and adiabatic

$$dS = 0$$

and

$$S = \text{constant}$$

A reversible adiabatic process is, therefore, an isentropic process.

Now

$$đQ_{\text{rev}} = TdS$$

or

$$Q_{\text{rev}} = \int_i^f TdS$$

The system is taken from i to f reversibly (Fig. 7.5). The area under the curve $\int_i^f TdS$ is equal to the heat transferred in the process.

For reversible isothermal heat transfer (Fig. 7.6), T = constant.

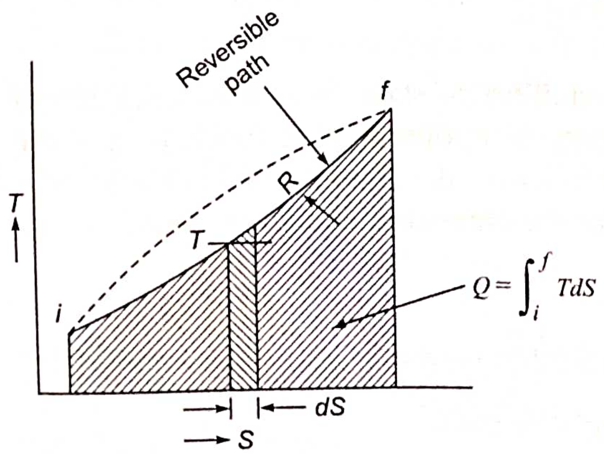

Fig. 7.5 *Area under a reversible path on the T-s plot represents heat transfer*

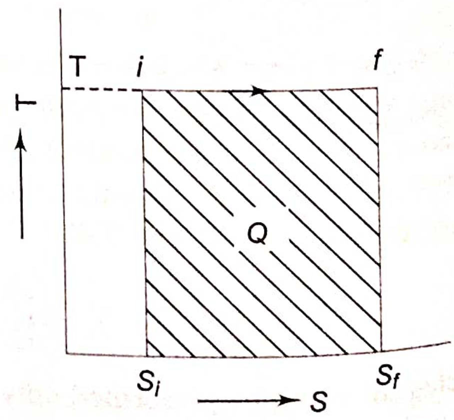

Fig. 7.6 *Reversible isothermal heat transfer*

$$\therefore \qquad Q_{\text{rev}} = T\int_i^f dS = T(S_f - S_i)$$

For a reversible adiabatic process, $dS = 0$, $S = C$ (Fig. 7.7).

The *Carnot cycle,* comprising two reversible isotherms and two reversible adiabatics, forms a rectangle in the T-S plane (Fig. 7.8). Process 4–1 represents reversible isothermal heat addition Q_1 to the system at T_1 from an external source, process 1–2 is the reversible adiabatic expansion of the system producing W_E amount of work, process 2–3 is the reversible isothermal heat rejection from the system to an external sink at T_2 and process 3–4 represents reversible adiabatic compression of the system consuming W_c amount of work. Area 1234 represents the net work output per cycle and the area under 4–1 indicates the quantity of heat added to the system Q_1.

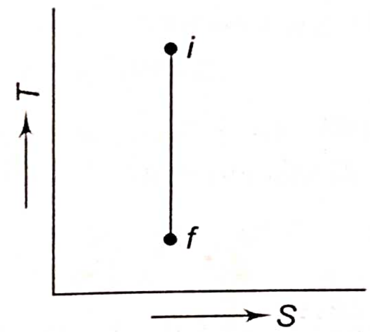

Fig. 7.7 *Reversible adiabatic is isentropic*

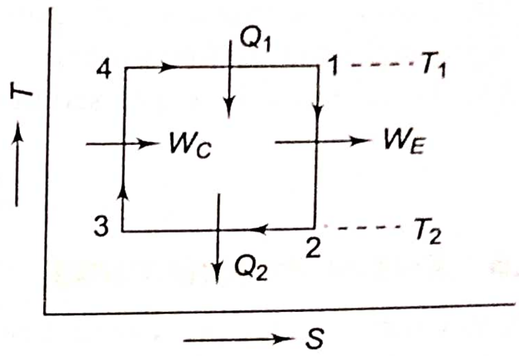

Fig. 7.8 *Carnot cycle*

$$\therefore \qquad \eta_{\text{Carnot}} = \frac{Q_1 - Q_2}{Q_1} = \frac{T_1 (S_1 - S_4) - T_2 (S_2 - S_3)}{T_1 (S_1 - S_4)}$$

$$= \frac{T_1 - T_2}{T_1} = 1 - \frac{T_2}{T_1}$$

and

$$W_{\text{net}} = Q_1 - Q_2 = (T_1 - T_2)(S_1 - S_4)$$

7.5 ENTROPY CHANGE IN AN IRREVERSIBLE PROCESS

LO7.5 Discuss the entropy change in any process

Let a closed system undergo an arbitrary process (reversible or irreversible) from State 1 to State 2, and let the cycle be completed thorough a reversible process 2-1 as shown in Fig. 7.9.

From the Clausius inequality, we have

$$\oint \frac{đQ}{T} \le 0$$

or

$$\int_1^2 \left(\frac{đQ}{T}\right) + \int_2^1 \left(\frac{đQ}{T}\right)_{\text{rev}} \le 0$$

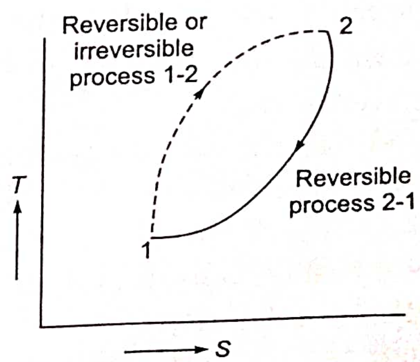

Fig. 7.9 *Entropy change in an irreversible process*

From the definition of entropy, we have $\int_2^1 \left(\frac{đQ}{T}\right)_{\text{rev}} = S_1 - S_2$. Therefore, the above equation reduces to

$$\int_1^2 \left(\frac{đQ}{T}\right) + S_1 - S_2 \le 0$$

or

$$S_2 - S_1 \ge \int_1^2 \left(\frac{đQ}{T}\right) \tag{7.14}$$

The equality sign holds good for a reversible process and the inequality sign for an irreversible process. It can be concluded from Eq. (7.14) that the change in entropy for an irreversible process undergone by a closed system is greater than the integral of $đQ/T$ evaluated for that process.

Equation (7.14) can be expressed in differential form as

$$dS \ge \frac{đQ}{T} \tag{7.15}$$

7.6 ENTROPY PRINCIPLE

LO7.6
Explain the principle of increase of entropy

For any infinitesimal process undergone by a system, we have the following equation for the total mass

$$dS \ge \frac{đQ}{T}$$

For an isolated system which does not undergo any energy interaction with the surroundings, $đQ = 0$.

Therefore, for an isolated system

$$dS_{\text{iso}} \ge 0 \tag{7.16}$$

For a reversible process,

$$dS_{\text{iso}} = 0$$

or

$$S = \text{constant}$$

For an irreversible process

$$dS_{\text{iso}} > 0$$

It is thus proved that *the entropy of an isolated system can never decrease*. It always increases and remains constant only when the process is reversible. This is known as the *principle of increase of entropy*, or simply the *entropy principle*. It is the quantitative general statement of second law from the macroscopic viewpoint.

An isolated system can always be formed by including any system and its surroundings within a single boundary (Fig. 7.10). Sometimes the original system which is then only a part of the isolated system is called a 'subsystem'.

The system and the surroundings together (the universe or the isolated system) include everything

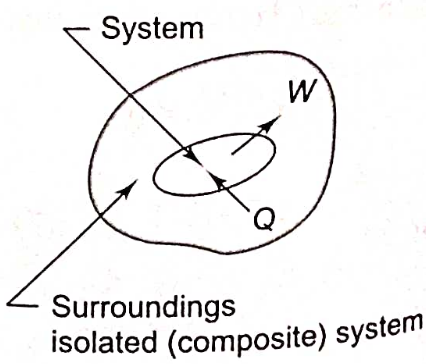

Fig. 7.10 *Isolated system*

which is affected by the process. For all possible processes that a system in the given surroundings can undergo

$$dS_{univ} \geq 0$$

or

$$dS_{sys} + dS_{surr} \geq 0 \tag{7.17}$$

Entropy may decrease locally at some region within the isolated system, but it must be compensated by a greater increase of entropy somewhere within the system so that the net effect of an irreversible process is an entropy increase of the whole system. This implies that entropy of a closed system may decrease during a process without violation of the entropy principle provided that the increase in entropy of the universe during that process is more than that of decrease in entropy of the system. The entropy increase of an isolated system is a measure of the extent of irreversibility of the process undergone by the system.

Rudolf Clausius summarised the first and second laws of thermodynamics in the following words:

1. *Die Energie der Welt ist konstant.*
2. *Die Entropie der Welt strebt einem Maximum zu.*

[1. The energy of the world (universe) is constant.
2. The entropy of the world tends towards a maximum.]

The entropy of an isolated system always increases and becomes maximum at the state of equilibrium. If the entropy of an isolated system varies with some parameter x, then there is a certain value of x_e which maximises the entropy $\left(\text{when } \dfrac{dS}{dx} = 0\right)$ and represents the equilibrium state (Fig. 7.11). The system is then said to exist at the peak of the entropy hill, and $dS = 0$. *When the system is at equilibrium, any conceivable change in entropy would be zero.*

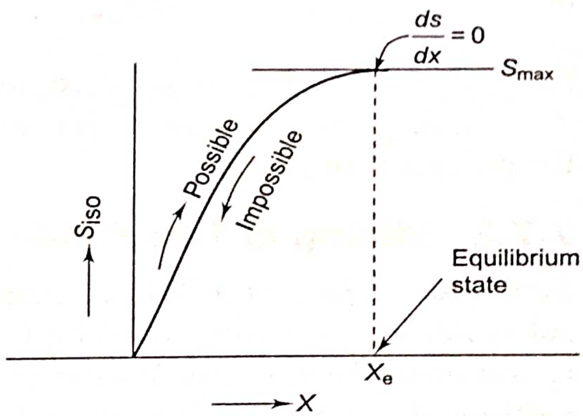

Fig. 7.11 *Equilibrium state of an isolated system*

7.7 APPLICATIONS OF ENTROPY PRINCIPLE

LO7.7 Describe the various applications of entropy principle in quantifying irreversibility of the process

The principle of increase of entropy is one of the most important laws of physical science. It is the quantitative statement of the second law of thermodynamics. Every irreversible process is accompanied by entropy increase of the universe, and this entropy increase quantifies the extent of irreversibility of the process. The higher the entropy increase of the universe the higher will be the irreversibility of the process. A few applications of the entropy principle are illustrated in the following sections.

7.7.1 Transfer of Heat through a Finite Temperature Difference

Let Q be the rate of heat transfer from reservoir A at T_1 to reservoir B at T_2, $T_1 > T_2$ (Fig. 7.12).

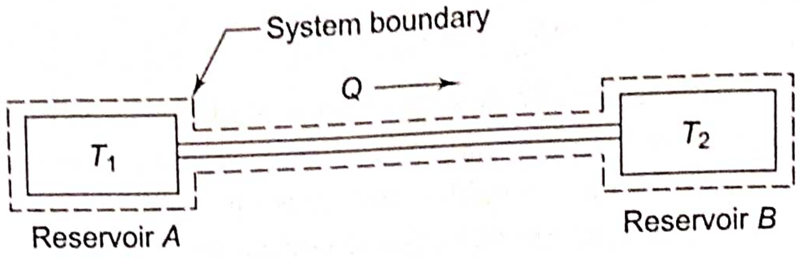

Fig. 7.12 *Heat transfer through a finite temperature difference*

For reservoir A, $\Delta S_A = -Q/T_1$. It is negative because heat Q flows out of the reservoir. For reservoir B, $\Delta S_B = +Q/T_2$. It is positive because heat flows into the reservoir. The rod connecting the reservoirs suffers no entropy change because once in the steady state, its coordinates do not change.

Therefore, for the isolated system comprising the reservoirs and the rod, and since entropy is an additive property

$$S = S_A + S_B$$

$$\Delta S_{\text{univ}} = \Delta S_A + \Delta S_B$$

or

$$\Delta S_{\text{univ}} = -\frac{Q}{T_1} + \frac{Q}{T_2} = Q \cdot \frac{T_1 - T_2}{T_2 T_2}$$

Since $T_1 > T_2$, ΔS_{univ} is positive, and the process is irreversible and possible. If $T_1 = T_2$, ΔS_{univ} is zero, and the process is reversible. If $T_1 < T_2$, ΔS_{univ} is negative and the process is impossible.

7.7.2 Mixing of Two Fluids

Subsystem 1 having a fluid of mass m_1, specific heat c_1 and temperature t_1 and subsystem 2 consisting of a fluid of mass m_2, specific heat c_2 and temperature t_2 comprise a composite system in an adiabatic enclosure (Fig. 7.13). When the partition is removed, the two fluids mix together, and at equilibrium let t_f be the final temperature, and $t_2 < t_f < t_1$. Since energy interaction is exclusively confined to the two fluids, the system being isolated

$$m_1 c_1 (t_1 - t_f) = m_2 c_2 (t_f - t_2)$$

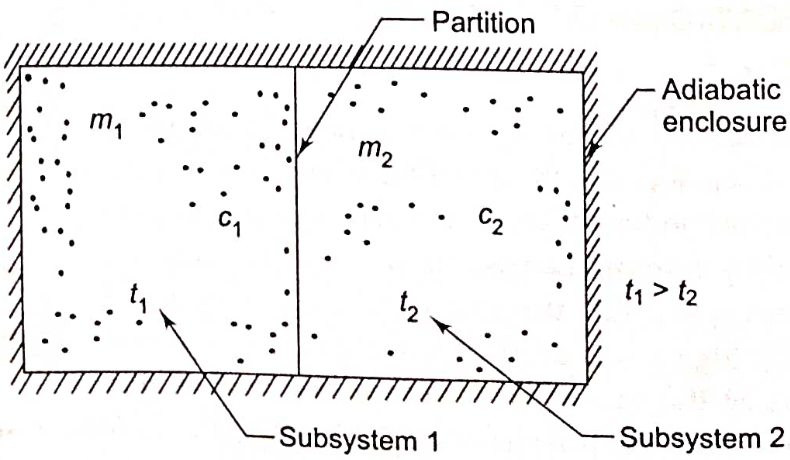

Fig. 7.13 *Mixing of two fluids*

$$\therefore \qquad t_f = \frac{m_1 c_1 t_1 + m_2 c_2 t_2}{m_1 c_1 + m_2 c_2}$$

Entropy change for the fluid in subsystem 1

$$\Delta S_1 = \int_{T_1}^{T_f} \frac{đQ_{\text{rev}}}{T} = \int_{T_i}^{T_f} \frac{m_1 c_1 dT}{T} = m_1 c_1 \ln \frac{T_f}{T_1}$$

$$= m_1 c_1 \ln \frac{t_f + 273}{t_1 + 273}$$

This will be negative, since $T_1 > T_f$

Entropy change for the fluid in subsystem 2

$$\Delta S_2 = \int_{T_2}^{T_f} \frac{m_2 c_2 dT}{T} = m_2 c_2 \ln \frac{T_f}{T_2} = m_2 c_2 \ln \frac{t_f + 273}{t_2 + 273}$$

This will be positive, since $T_2 < T_f$

$$\therefore \qquad \Delta S_{\text{univ}} = \Delta S_1 + \Delta S_2$$

$$= m_1 c_1 \ln \frac{T_f}{T_1} + m_2 c_2 \ln \frac{T_f}{T_2}$$

ΔS_{univ} will be positive definite, and the mixing process is irreversible.

Although the mixing process is irreversible, to evaluate the entropy change for the subsystems, the irreversible path was replaced by a reversible path on which the integration was performed.

If $m_1 = m_2 = m$ and $c_1 = c_2 = c$.

$$\Delta S_{\text{univ}} = mc \ln \frac{T_f^2}{T_1 \cdot T_2}$$

and

$$T_f = \frac{m_1 c_1 T_1 + m_2 c_2 T_2}{m_1 c_1 + m_2 c_2} = \frac{T_1 + T_2}{2}$$

$$\therefore \qquad \Delta S_{\text{univ}} = 2mc \ln \frac{(T_1 + T_2)/2}{\sqrt{T_1 \cdot T_2}}$$

This is always positive, since the arithmetic mean of any two numbers is always greater than their geometric mean. This can also be proved geometrically. Let a semi-circle be drawn with $(T_1 + T_2)$ as diameter (Fig. 7.14).

Here, $AB = T_1$, $BC = T_2$ and $OE = (T_1 + T_2)/2$. It is known that $(DB)^2 = AB \cdot BC = T_1 T_2$.

$$\therefore \qquad DB = \sqrt{T_1 T_2}$$

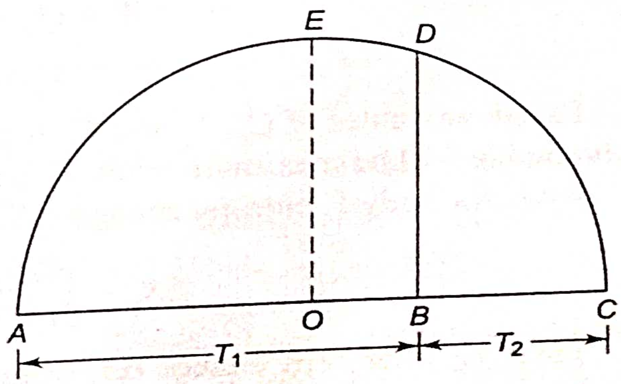

Fig. 7.14 *Geometrical proof to show that g.m. < a.m.*

Now, $OE > DB$

$$\frac{T_1 + T_2}{2} > \sqrt{T_1 T_2}$$

7.7.3 Maximum Work Obtainable from Two Finite Bodies at Temperatures T_1 and T_2

Let us consider two identical finite bodies of constant heat capacity at temperatures T_1 and T_2, respectively, T_1 being higher than T_2. If the two bodies are merely brought together into thermal contact, delivering no work, the final temperature T_f reached would be the maximum

$$T_f = \frac{T_1 + T_2}{2}$$

If a heat engine is operated between the two bodies acting as thermal energy reservoirs (Fig. 7.15), part of the heat withdrawn from body 1 is converted to work W by the heat engine, and the remainder is rejected to body 2. The lowest attainable final temperature T_f corresponds to the delivery of the largest possible amount of work, and is associated with a reversible process.

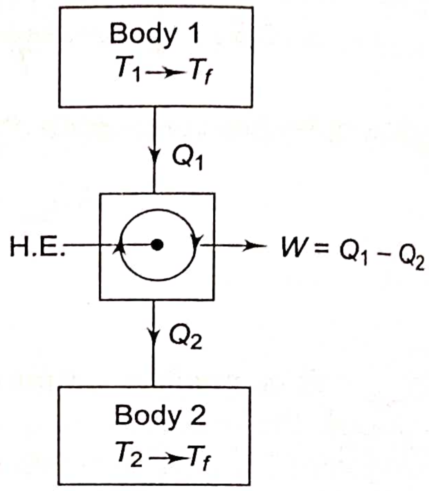

Fig. 7.15 *Maximum work obtainable from two finite bodies*

As work is delivered by the heat engine, the temperature of body 1 will be decreasing and that of body 2 will be increasing. When both the bodies attain the final temperature T_f, the heat engine will stop operating. Let the bodies remain at constant pressure and undergo no change of phase.

Total heat withdrawn from body 1

$$Q_1 = C_p (T_1 - T_f)$$

where C_p is the heat capacity of the two bodies at constant pressure

Total heat rejected to body 2

$$Q_2 = C_p (T_f - T_2)$$

∴ amount of total work delivered by the heat engine

$$W = Q_1 - Q_2$$

$$= C_p (T_1 + T_2 - 2T_f) \tag{7.18}$$

For given values of C_p, T_1 and T_2, the magnitude of work W depends on T_f. Work obtainable will be maximum when T_f is minimum.

Now, for body 1, entropy change ΔS_1 is given by

$$\Delta S_1 = \int_{T_1}^{T_f} C_p \frac{dT}{T} = C_p \ln \frac{T_f}{T_1}$$

For body 2, entropy change ΔS_2 would be

$$\Delta S_2 = \int_{T_2}^{T_f} C_p \frac{dT}{T} = C_p \ln \frac{T_f}{T_2}$$

Since the working fluid operating in the heat engine cycle does not undergo any entropy change, ΔS of the working fluid in heat engine $\oint dS = 0$. Applying the entropy principle

$$\Delta S_{\text{univ}} \geq 0$$

$$\therefore \quad C_p \ln \frac{T_f}{T_1} + C_p \ln \frac{T_f}{T_2} \geq 0$$

$$C_p \ln \frac{T_f^2}{T_1 T_2} \geq 0 \tag{7.19}$$

From Eq. (7.19), for T_f to be a minimum

$$C_p \ln \frac{T_f^2}{T_1 \cdot T_2} = 0$$

or

$$\ln \frac{T_f^2}{T_1 T_2} = 0 = \ln 1$$

$$\therefore \quad T_f = \sqrt{T_1 \cdot T_2} \tag{7.20}$$

For W to be a maximum, T_f will be $\sqrt{T_1 T_2}$. From Eq. (7.19)

$$W_{\max} = C_p\,(T_1 + T_2 - 2\sqrt{T_1 T_2}\,) = C_p(\sqrt{T_1} - \sqrt{T_2})^2$$

The final temperatures of the two bodies, initially at T_1 and T_2, can range from $(T_1 + T_2)/2$ with no delivery of work to $\sqrt{T_1 T_2}$ with maximum delivery of work.

7.7.4 Maximum Work Obtainable from a Finite Body and a TER

Let one of the bodies considered in the previous section be a thermal energy reservoir. The finite body has a thermal capacity C_p and is at temperature T and the TER is at temperature T_0, such that $T > T_0$. Let a heat engine operate between the two (Fig. 7.16). As heat is withdrawn from the body, its temperature decreases. The temperature of the TER would, however, remain unchanged at T_0. The engine would stop working when the temperature of the body reaches T_0. During that period, the amount of work delivered is W, and the heat rejected to the TER is $(Q - W)$. Then

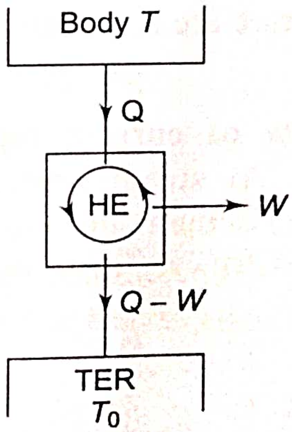

Fig. 7.16 *Maximum work obtainable when one of the bodies is a TER*

$$\Delta S_{\text{Body}} = \int_T^{T_0} C_p \frac{dT}{T} = C_p \ln \frac{T_0}{T}$$

$$\Delta S_{\text{HE}} = \oint dS = 0$$

$$Q = C_p\,(T - T_0)$$

$$\Delta S_{\text{TER}} = \frac{Q - W}{T_0}$$

$$\therefore \qquad \Delta S_{\text{univ}} = C_p \ln \frac{T_0}{T} + \frac{Q - W}{T_0}$$

By the entropy principle,

$$\Delta S_{\text{univ}} \geq 0$$

$$C_p \ln \frac{T_0}{T} + \frac{Q - W}{T_0} \geq 0$$

or
$$C_p \ln \frac{T_0}{T} \geq \frac{W - Q}{T_0}$$

or
$$\frac{W - Q}{T_0} \leq C_p \ln \frac{T_0}{T}$$

or
$$W \leq Q + T_0 C_p \ln \frac{T_0}{T}$$

$$\therefore \qquad W_{\max} = Q + T_0 C_p \ln \frac{T_0}{T}$$

or,
$$W_{\max} = C_p \left[(T - T_0) - T_0 \ln \frac{T}{T_0} \right] \tag{7.21}$$

It is called the *exergy* of the finite body at temperature T.

7.7.5 Processes Exhibiting External Mechanical Irreversibility

1. Isothermal dissipation of work Let us consider the isothermal dissipation of work through a system into the internal energy of a reservoir, as in the flow of an electric current I through a resistor in contact with a reservoir (Fig. 7.17). At steady state, the internal energy of the resistor and hence its temperature are constant. So, by first law

$$\dot{W} = \dot{Q}$$

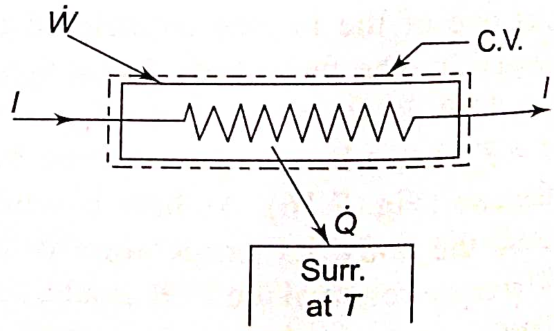

Fig. 7.17 *External mechanical irreversibility*

The flow of current represents work transfer. At steady state, the work is dissipated isothermally into heat transfer to the surroundings. Since the surroundings absorb Q units of heat at temperature T,

$$\dot{\Delta S}_{\text{surr}} = \frac{\dot{Q}}{T} = \frac{\dot{W}}{T}$$

At steady state, $\dot{\Delta S}_{\text{sys}} = 0$

$$\therefore \qquad \dot{\Delta S}_{\text{univ}} = \dot{\Delta S}_{\text{sys}} + \dot{\Delta S}_{\text{surr}} = \frac{\dot{W}}{T} \tag{7.22}$$

The irreversible process is thus accompanied by an entropy increase of the universe.

2. Adiabatic dissipation of work Let W be the stirring work supplied to a viscous thermally insulated liquid, which is dissipated adiabatically into internal energy increase of the liquid, the temperature of which increases from T_i to T_f (Fig. 7.18). Since there is no flow of heat to or from the surroundings,

$$\Delta S_{\text{surr}} = 0$$

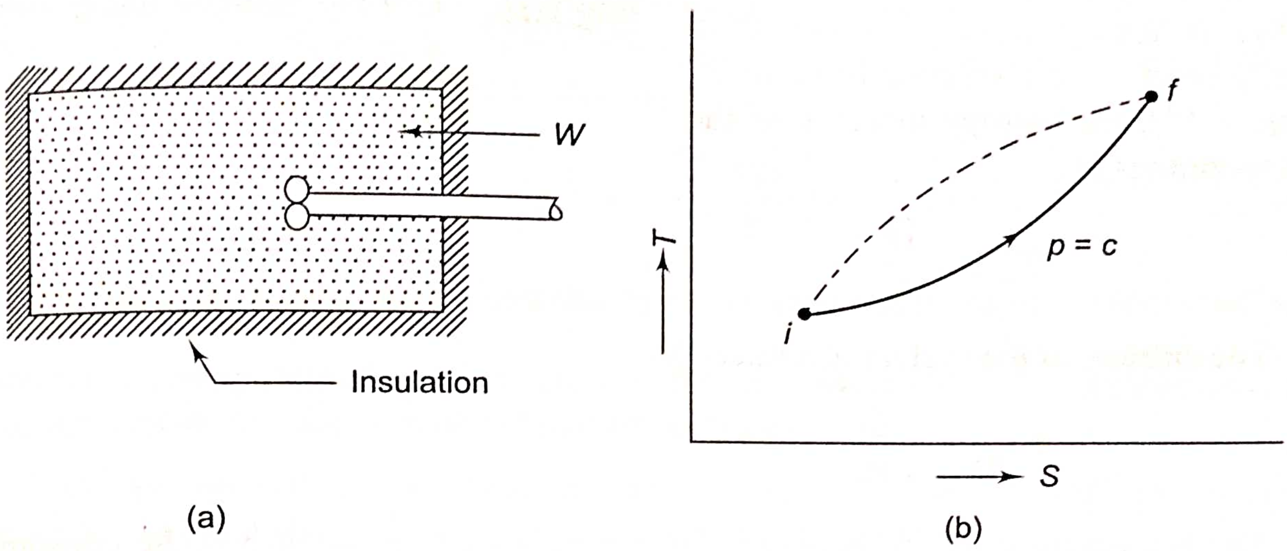

Fig. 7.18 *Adiabatic dissipation of work*

To calculate the entropy change of the system, the original irreversible path (dotted line) must be replaced by a reversible one between the same end states, i and f. Let us replace the irreversible performance of work by a reversible isobaric flow of heat from a series of reservoirs ranging from T_i to T_f to cause the same change in the state of the system. The entropy change of the system will be

$$\Delta S_{\text{sys}} = \underset{R}{\int_i^f} \frac{đQ}{T} = \underset{R}{\int_i^f} \frac{C_p dT}{T} = C_p \ln \frac{T_f}{T_i}$$

where C_p is the heat capacity of the liquid.

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = C_p \ln \frac{T_f}{T_i} \tag{7.23}$$

which is positive.

7.8 ENTROPY TRANSFER WITH HEAT FLOW

LO7.8 Analyzing entropy transfer with flow of heat and not with work

Since $dS = \dfrac{đQ_{\text{rev}}}{T}$, when heat is added to a system, $đQ$ is positive, and the entropy of the system increases. When heat is removed from the system, $đQ$ is negative, and the entropy of the system decreases.

Heat transferred to the system of fixed mass increases the internal energy of the system, as a result of which the molecules (of a gas) move with higher kinetic energy and collide more frequently, and so the disorder in the system increases. Heat is thus regarded as disorganised or disordered energy transfer which increases molecular chaos (*see Section 7.14*). If heat Q flows reversibly from the system to the surroundings at T_0 (Fig. 7.19), the entropy increase of the surroundings is

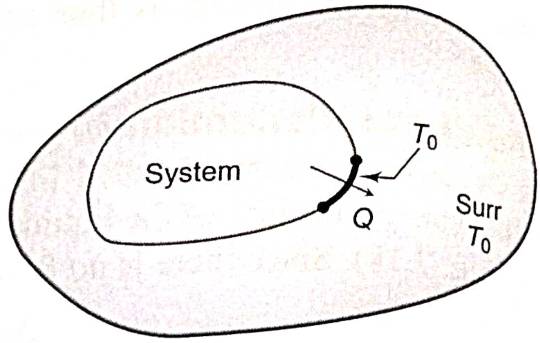

Fig. 7.19 *Entropy transfer along with heat flow*

$$\Delta S_{\text{surr}} = \frac{Q}{T_0}$$

The entropy of the system is reduced by

$$\Delta S_{\text{sys}} = -\frac{Q}{T_0}$$

The temperature of the boundary where heat transfer occurs is the constant temperature T_0. It may be said that the system has lost entropy to the surroundings. Alternatively, one may state that the surroundings have gained entropy from the system. Therefore, there is *entropy transfer* from the system to the surroundings along with heat flow. In other words, since the heat inflow increases the molecular disorder, there is flow of disorder along with heat.

On the other hand, *there is no entropy transfer associated with work.* In Fig. 7.20, the system delivers work to a flywheel where energy is stored in a fully recoverable form. The flywheel molecules are simply put into rotation around the axis in a perfectly organised manner, and there is no dissipation and hence no entropy increase of the flywheel. The same can be said about work transfer in the compression of a spring or in the raising of a weight by a certain height.

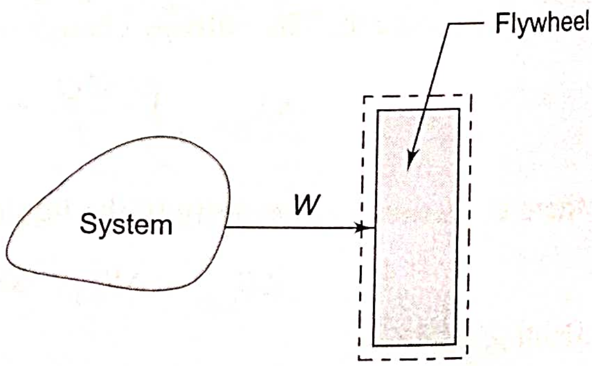

Fig. 7.20 *No entropy transfer along with work transfer*

There is thus no entropy transfer along with work. If work is dissipated adiabatically into internal energy increase of the system (Subsection 7.7.5), there is an entropy increase in the system, but there is as such no entropy transfer to it.

Thus, we can distinguish work and heat in this way: 'Work' is energy transfer without associated entropy transfer and 'Heat' is energy transfer with associated entropy transfer.

7.9 ENTROPY GENERATION IN A CLOSED SYSTEM

LO 7.9
Examine possible ways of entropy increase in closed and open system

The change in entropy during a process undergone by a closed system can be written using Eq. (7.14) as

$$S_2 - S_1 \geq \int_1^2 \left(\frac{đQ}{T} \right)$$

The inequality sign in the above equation implies that the change in entropy for an irreversible process undergone by a closed system is greater than the integral of $đQ/T$ evaluated for that process and hence some entropy is generated during this process. Thus, the above equation can be written as

$$S_2 - S_1 = \int_1^2 \left(\frac{đQ}{T} \right) + S_{gen} \tag{7.24}$$

where S_{gen} is the entropy generation during the process due to internal irreversibilities present in the system. Thus, we can say the expression $dS - \frac{đQ}{T}$ represents the entropy generation due to internal irreversibilities.

It can be concluded from the above equation that the change of entropy during a process undergone by a closed system takes place due to two reasons:

1. By heat interaction in which there is entropy transfer.
2. By internal irreversibilities or dissipative effects in which work (or K.E.) is dissipated into internal energy and in this way internal energy of the system increases.

Note that the entropy transfer due to heat interaction for a system may be positive or negative depending on whether heat is transferred to the system or from the system. On the contrary, the entropy generation S_{gen} is always a positive quantity or zero in the limit if the process is reversible. The value of S_{gen} depends on the process and hence it is not a property of the system.

It may so happen that in a process (e.g., the expansion of a hot fluid in a turbine) the entropy decrease of the system due to heat loss to the surroundings $\left(-\int \frac{đQ}{T} \right)$ is equal to the entropy increase of the system due to internal irreversibilities such as friction, etc. (S_{gen}), in which case the entropy of the system before and after the process will remain the same $\left(\int dS = 0 \right)$. *Therefore, an isentropic process need not be adiabatic or reversible. But if the isentropic process is reversible, it must be adiabatic. Also, if the isentropic process is adiabatic, it cannot but be reversible. An adiabatic process need not be isentropic, since entropy can also increase due to friction etc. But if the process is adiabatic and reversible, it must be isentropic.*

For an infinitesimal reversible process by a closed system involving only pdV-work,

$$đQ_R = dU_R + pdV$$

If the process is irreversible,

$$đQ_I = dU_I + đW$$

Since U is a property,

$$dU_R = dU_I$$

$$\therefore \quad đQ_R - pdV = đQ_I - đW$$

or

$$\left(\frac{đQ}{T}\right)_R = \left(\frac{dQ}{T}\right)_I + \frac{pdV - đW}{T} \tag{7.25}$$

The difference $(pdV - đW)$ indicates the work that is lost due to irreversibility, and is called the *lost work* $d(LW)$, which approaches zero as the process approaches reversibility as a limit. Equation (7.26) can be expressed in the form

$$dS = \frac{đQ}{T} + \frac{d(LW)}{T} \tag{7.26}$$

From the above equation it can be physically interpreted that the entropy generation S_{gen} is associated with the lost work due to internal irreversibility present in the system. The lost work is always positive and hence the entropy generation is a positive quantity.

In any process executed by a system, energy is always conserved, but entropy is produced internally. For any process between equilibrium states 1 and 2 (Fig. 7.21), the first law can be written as follows:

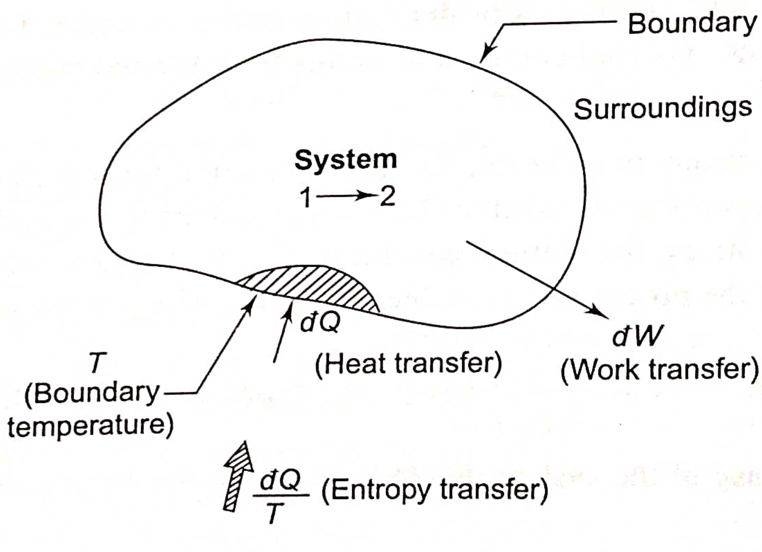

Fig. 7.21 *Schematic of a closed system interacting with its surroundings*

$$\int_1^2 đQ - \int_1^2 đW = E_2 - E_1$$

or

Energy transfer   Energy change

$$Q_{1-2} = E_2 - E_1 + W_{1-2}$$

By the second law,

$$S_2 - S_1 \geq \int_1^2 \frac{đQ}{T}$$

7.10 ENTROPY GENERATION IN AN OPEN SYSTEM

In an open system, there is transfer of three quantities: mass, energy and entropy. The control surface can have one or more openings for mass transfer (Fig. 7.22). It is rigid, and there is shaft work transfer across it.

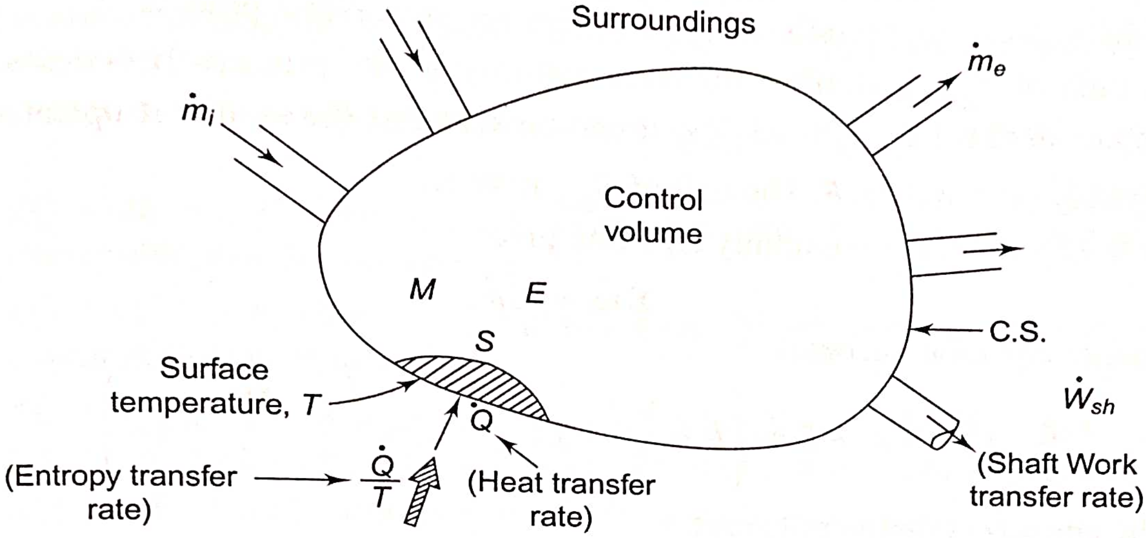

Fig. 7.22 *Schematic of an open system and its interaction with surroundings*

The *continuity equation* gives

$$\underset{i}{\Sigma}\dot{m}_1 - \underset{e}{\Sigma}\dot{m}_e = \frac{\partial M}{\partial \tau} \tag{7.27}$$

net mass transfer rate   rate of mass accumulation in the CV

The *energy equation* gives·

$$\underset{i}{\Sigma}\dot{m}_1\left(h + \frac{\mathbf{V}^2}{2} + gZ\right)_i - \underset{e}{\Sigma}\dot{m}_e\left(h + \frac{\mathbf{V}^2}{2} + gZ\right)_e + \dot{Q} - \dot{W}_{\text{sh}} = \frac{\partial E}{\partial \tau} \tag{7.28}$$

net rate of energy transfer   rate of energy accumulation in the CV

The *second law inequality* or the entropy principle gives

$$\underset{i}{\Sigma}\dot{m}_i S_i - \underset{e}{\Sigma}\dot{m}_e S_e + \frac{\dot{Q}}{T} \leq \frac{\partial S}{\partial \tau} \tag{7.29}$$

net rate of entropy transfer   rate of increase of entropy of the CV

Here $\dot{Q}$ represents the rate of heat transfer at the location of the boundary where the instantaneous temperature is T. The ratio $\dot{Q}/T$ accounts for the entropy transfer along with heat. The terms $\dot{m}_i S_i$ and $\dot{m}_e S_e$ account, respectively, for rates of entropy transfer into and out of the CV accompanying mass flow. The rate of entropy increase of the control volume exceeds, or is equal to, the net rate of entropy transfer into it. The difference is the entropy generated within the control volume due to irreversibility. Hence, the rate of entropy generation is given by

$$\dot{S}_{\text{gen}} = \frac{\partial S}{\partial \tau} - \sum_i \dot{m}_i S_i + \sum_e \dot{m}_e S_e - \frac{\dot{Q}}{T} \tag{7.30}$$

By the second law,

$$\dot{S}_{\text{gen}} \geq 0$$

If the process is reversible, $\dot{S}_{\text{gen}} = 0$. For an irreversible process, $\dot{S}_{\text{gen}} > 0$. The magnitude of $\dot{S}_{\text{gen}}$ quantifies the irreversibility of the process. If systems A and B operate so that $(\dot{S}_{\text{gen}})_A > (\dot{S}_{\text{gen}})_B$ it can be said that the system A operates more irreversibly than system B. The unit of $\dot{S}_{\text{gen}}$ is W/K.

At *steady state*, the continuity equation gives

$$\sum_i \dot{m}_i = \sum_e \dot{m}_e \tag{7.31}$$

the energy equation becomes

$$0 = \dot{Q} - \dot{W}_{\text{sh}} + \sum_i \dot{m}_i \left(h + \frac{\mathbf{V}^2}{2} + gZ \right)_i - \sum_e \dot{m}_e \left(h + \frac{\mathbf{V}^2}{2} + gZ \right)_e \tag{7.32}$$

and the entropy equation reduces to

$$0 = \frac{\dot{Q}}{T} + \sum_i \dot{m}_i S_i - \sum_e \dot{m}_e s_e + \dot{S}_{\text{gen}} \tag{7.33}$$

These equations often must be solved simultaneously, together with appropriate property relations.

Mass and energy are conserved quantities, but entropy is not generally conserved. The rate at which entropy is transferred out must exceed the rate at which entropy enters the CV, the difference being the rate of entropy generated within the CV owing to irreversibilities.

For one-inlet and one-exit control volumes, the entropy equation becomes

$$0 = \frac{\dot{Q}}{T} + \dot{m}(s_1 - s_2) + \dot{S}_{\text{gen}}$$

$$\therefore \quad s_2 - s_1 = \frac{1}{\dot{m}} \left(\frac{\dot{Q}}{T} \right) + \frac{\dot{S}_{\text{gen}}}{\dot{m}} \tag{7.34}$$

7.11 COMBINED CONSEQUENCES OF FIRST AND SECOND LAWS

LO 7.10 Express combined consequences of First and Second laws

The heat transfer during a reversible process can be expressed using the second law as

$$đQ_{\text{rev}} = TdS$$

and by the first law, for a closed non-flow system involving only *pdV* work,

$$đQ_{rev} = dU + pdV$$

Combing the above two equations, we get

$$TdS = dU + pdV \tag{7.35}$$

Again the enthalpy

$$H = U + pV$$

or $$dH = dU + pdV + Vdp$$

or $$dH = TdS + Vdp$$

or $$TdS = dH - Vdp \tag{7.36}$$

Equations (7.35) and (7.36) are the thermodynamic equations relating the properties of the system and are used to estimate the entropy change of a system in a process between two states.

Let us now examine the following equations as obtained from the first and second laws:

1. $đQ = dE + đW$; This equation holds good for any process, reversible or irreversible, and for any system.
2. $đQ = dU + đW$; This equation holds good for any process undergone by a closed stationary system.
3. $đQ = dU + pdV$; This equation holds good for a reversible (quasi-static) process undergone by a closed system when only pdV-work is present.
4. $đQ = TdS$; This equation is true only for a reversible process.
5. $TdS = dU + pdV$; This equation holds good for any process reversible or irreversible, undergone by a closed system involving only pdV -work, since it is a relation among properties, which are independent of the path.
6. $TdS = dH - Vdp$; This equation also relates only the properties of the closed system. There is no path function term in the equation. Hence, the equation holds good for any process when only pdV-work is present.

The use of the term 'irreversible process' is not proper, since no irreversible path or process can be plotted on thermodynamic coordinates. It is more logical to state that 'the change of state is irreversible' rather than say 'it is an irreversible process'. A natural process which is inherently irreversible is indicated by a dotted line connecting the initial and final states, both of which are in equilibrium. The dotted line has no other meaning, since it can be drawn in any way. To determine the entropy change for a real process, a known reversible path is made to connect the two end states, and integration is performed on this path using either Eq. 5 or Eq. 6, as given above. Therefore, the entropy change of a system between two identifiable equilibrium states is the same whether the intervening process is reversible or the change of state is irreversible.

7.12 REVERSIBLE ADIABATIC WORK IN A STEADY FLOW SYSTEM

LO 7.11
Obtain reversible work done in a steady flow system

In the differential form, the steady flow energy equation per unit mass is given by Eq. (5.11),

$$đQ = dh + \mathbf{V}d\mathbf{V} + gdZ + đW_x$$

For a reversible process, $đQ = Tds$

$$\therefore \qquad Tds = dh + \mathbf{V}d\mathbf{V} + gdZ + đW_x \tag{7.37}$$

Using the property relation, Eq. (7.36), per unit mass

$$Tds = dh - vdp$$

In Eq. (7.37), we have

$$-v\delta p = \mathbf{V}d\mathbf{V} + gdZ + đW_x \tag{7.38}$$

On integration

$$-\int_1^2 vdp = \Delta\frac{\mathbf{V}^2}{2} + g\Delta Z + W_x \tag{7.39}$$

If the changes in K.E. and P.E. are neglected, Eq. (7.39) reduces to

$$W_x = -\int_1^2 vdp \tag{7.40}$$

If $đQ = 0$, implying $dS = 0$, the property relation gives

$$dh = vdp$$

or

$$h_2 - h_1 = \int_1^2 vdp \tag{7.41}$$

From Eqs. (7.40) and (7.41),

$$W_x = h_1 - h_2 = -\int_1^2 vdp \tag{7.42}$$

The integral $-\int_1^2 vdp$ represents an area on the p–v plane (Fig. 7.23). To make the integration, one must have a relation between p and v such as pv^n = constant.

$$\therefore \qquad W_{1-2} = h_1 - h_2 = -\int_1^2 vdp$$

$$= \text{area } 12\,ab\,1$$

Equation (7.42) holds good for a steady flow work-producing machine like an engine or turbine as well as for a work-absorbing machine like a pump or a compressor, when the fluid undergoes reversible adiabatic expansion or compression.

Figure 7.23 shows some steady flow machines where the work done is $-\int_1^2 vdp$

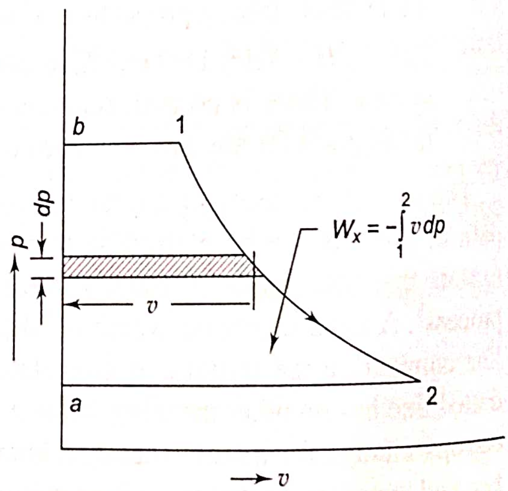

Fig. 7.23 *Reversible steady flow work interaction*

It may be noted that for a closed stationary system like a gas confined in a piston-cylinder machine (Fig. 7.24a), the reversible work done would be

$$W_{1-2} = \int_1^2 pdV = \text{Area } 12\,cd\,1$$

The reversible work done by a steady flow system (Fig. 7.24b) would be

$$W_{1-2} = -\int_1^2 v\,dp = \text{Area } 12\,ab\,1$$

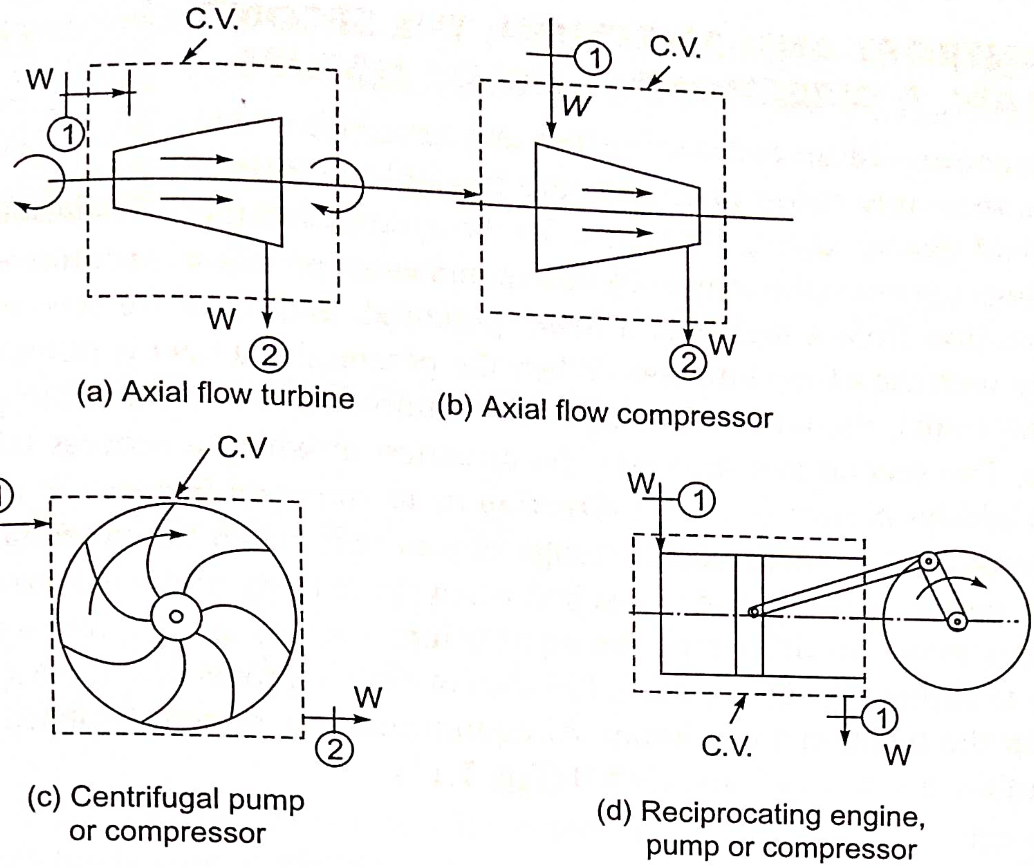

Fig. 7.24 *Work done by steady flow machines,* $-\int_1^2 v\,dp$

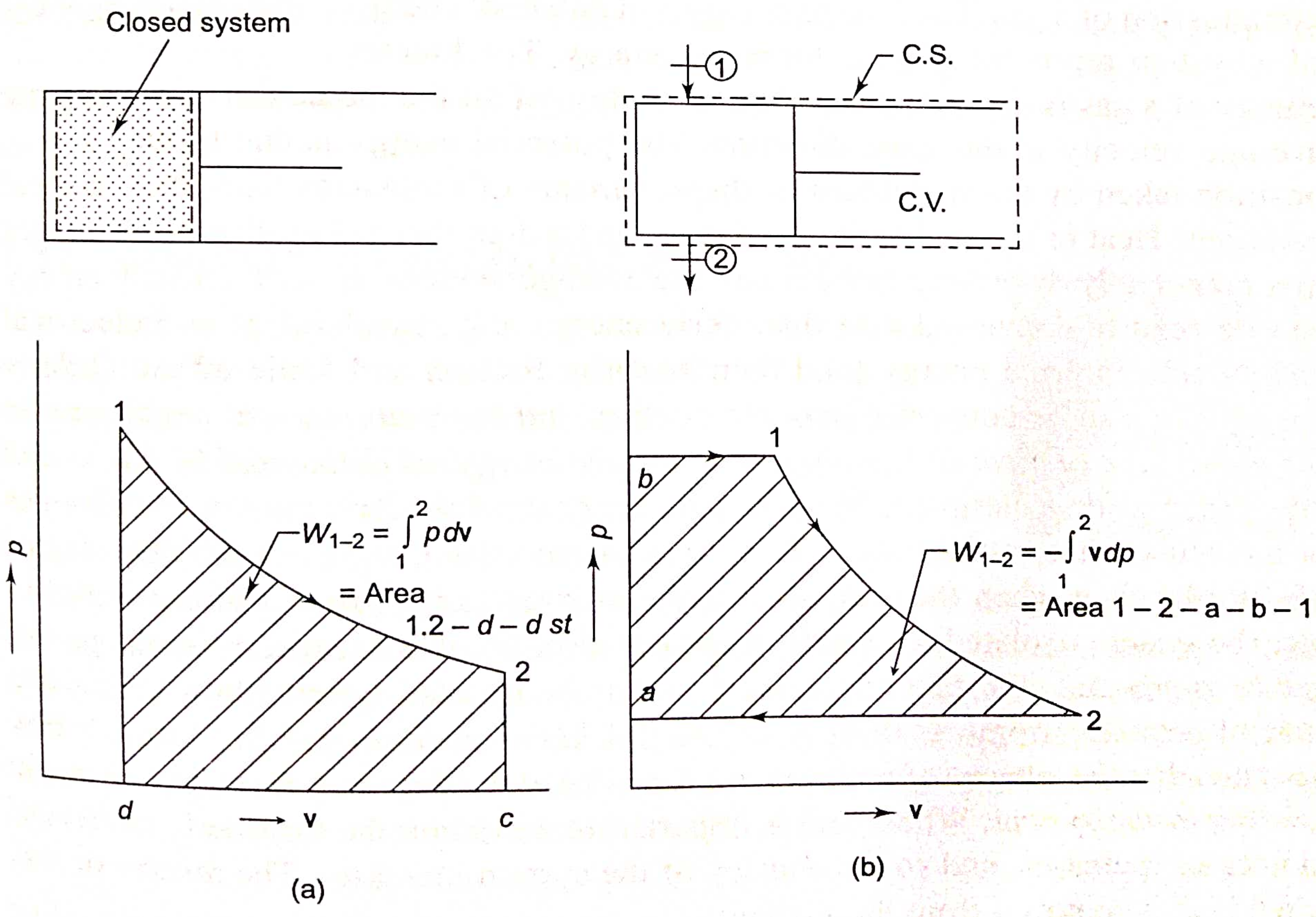

Fig. 7.25 *Reversible work transfer in (a) a closed system, (b) a steady flow system*

7.13 ENTROPY AND DIRECTION: THE SECOND LAW, A DIRECTIONAL LAW OF NATURE

LO 7.12 Express entropy and its dependence on direction

Since the entropy of an isolated system can never decrease, it follows that only those processes are possible in nature which would give an entropy increase for the system and the surroundings together (the universe). All spontaneous processes in nature occur only in one direction from a higher to a lower potential, and these are accompanied by an entropy increase of the universe. When the potential gradient is infinitesimal (or zero in the limit), the entropy change of the universe is zero, and the process is reversible. The second law indicates the direction in which a process takes place. *A process always occurs in such a direction as to cause an increase in the entropy of the universe.* The macroscopic change ceases only when the potential gradient disappears and the equilibrium is reached when the entropy of the universe assumes a maximum value. To determine the equilibrium state of an isolated system, it is necessary to express the entropy as a function of certain properties of the system and then render the function a maximum. At equilibrium, the system (isolated) exists at the peak of the entropy hill, and $dS = 0$ (Fig. 7.11).

7.14 ENTROPY AND DISORDER

LO 7.13 Correlate entropy with disorderness

Work is a macroscopic concept. Work involves order or the orderly motion of molecules, as in the expansion or compression of a gas. The kinetic energy and potential energy of a system represent orderly forms of energy. The kinetic energy of a gas is due to the coordinated motion of all the molecules with the same average velocity in the same direction. The potential energy is due to the vantage position taken by the molecules or displacements of molecules from their normal positions. Heat or thermal energy is due to the random thermal motion of molecules in a completely disorderly fashion and the average velocity is zero. Orderly energy can be readily converted into disorderly energy, e.g., mechanical and electrical energy into internal energy (and then heat) by friction and Joule effect. Orderly energy can also be converted into one another. *But there are natural limitations on the conversion of disorderly energy into orderly energy, as delineated by the second law.* When work is dissipated into internal energy, the disorderly motion of molecules or molecular chaos is increased. Two gases, when mixed, represent a higher degree of disorder than when they are separated. An irreversible process always tends to take the system (isolated) to a state of greater disorder. It is a tendency on the part of nature to proceed to a state of greater disorder. An isolated system always tends to a state of greater entropy. So there is a close link between entropy and disorder. It may be stated that *the entropy of a system is a measure of the degree of molecular disorder existing in the system.* When heat is imparted to a system, the disorderly motion of molecules increases, and so the entropy of the system increases. The reverse occurs when heat is removed from the system.

Ludwig Boltzmann (1877) introduced statistical concepts to define disorder by attaching to each state a *thermodynamic probability*, expressed by the quantity W, which is greater the more disordered the state is. The increase of entropy implies that the system proceeds by itself from one state to another with a higher thermodynamic probability (or disorder number). An irreversible process goes on until the *most probable state* (equilibrium state when W is maximum) corresponding to the maximum value of entropy is reached. Boltzmann assumed a functional relation between S and W. While entropy is additive, probability is multiplicative. If the two parts A and B of a system in equilibrium are considered, the entropy is the sum

$$S = S_A + S_B$$

and the thermodynamic probability is the product

$$W = W_A \cdot W_B$$

Again, $S = S(W)$, $S_A = S(W_A)$ and $S_B = S(W_B)$

$$\therefore \qquad S(W) = S(W_A W_B) = S(W_A) + S(W_B)$$

which is a well-known functional equation for the logarithm. Thus the famous relation is reached

$$S = K \ln W$$

where K is a constant, known as Boltzmann constant. This is engraved upon Boltzmann's tombstone in Vienna.

When $W = 1$, which represents the greatest order, $S = 0$. This occurs only at $T = 0\,K$. This state cannot be reached in a finite number of operations. This is the Nernst Simon statement of third law of thermodynamics. In the case of a gas, W increases due to an increase in volume V or temperature T. In the reversible adiabatic expansion of a gas, the increase in disorder due to an increase in volume is just compensated by the decrease in disorder due to a decrease in temperature, so that the disorder number or entropy remains constant.

7.15 CONCEPT OF ABSOLUTE ENTROPY

LO 7.14
Discuss the concepts of absolute entropy and postulatory thermodynamics

The entropy of a pure crystalline substance at absolute zero temperature is a state of perfect order ($W = 1$) and zero entropy. Since absolute zero temperature is not attainable in a finite number of operations (Section 6.16 in Chapter 6), the absolute zero entropy value is not attainable. It is the Nernst–Simon statement of third law of thermodynamics.

It is important to note that one is interested only in the amount by which the entropy of the system changes in going from an initial to final state, and not in the value of absolute entropy. In cases where it is necessary, a zero value of entropy of the system at an arbitrarily chosen standard state is assigned, and the entropy changes are calculated with reference to this standard state. At triple point of water, the entropy (also internal energy) is arbitrarily chosen to be zero.

7.16 POSTULATORY THERMODYNAMICS

The property 'entropy' plays the central role in thermodynamics. In the *classical approach*, as followed in this book, entropy is introduced via the concept of the heat engine. It follows the way in which the subject of thermodynamics developed historically, mainly through the contributions of Sadi Carnot, James Prescott Joule, William Thomson (Lord Kelvin), Rudolf Clausius, Max Planck and Josiah Willard Gibbs.

In the *postulatory approach*, as developed by HB Callen (see Reference), entropy is introduced at the beginning. The development of the subject has been based on four postulates. Postulate I defines the equilibrium state. Postulate II introduces the property 'entropy' which is rendered maximum at the final equilibrium state. Postulate III refers to the additive nature of entropy which is a monotonically increasing function of energy. Postulate IV mentions that the entropy of any system vanishes at the absolute zero of temperature. With the help of these postulates, the conditions of equilibrium under different constraints have been developed.